

Toxicity Detection for Generative AI Systems: A Survey of Evidence and Governance Validity

KUN XU, Nanjing University of Aeronautics and Astronautics, China

YUSHU ZHANG[✉], Jiangxi University of Finance and Economics, China

XIANGLI XIAO, Jiangxi University of Finance and Economics, China

SHUREN QI, City University of Hong Kong, China

TAO WANG, Nanjing University of Aeronautics and Astronautics, China

BARBARA CARMINATI, University of Insubria, Italy

ELENA FERRARI, University of Insubria, Italy

YUMING FANG, Jiangxi University of Finance and Economics, China

Toxicity detection is essential for limiting harmful exposure without suppressing legitimate expression. Generative systems complicate detection through interaction, multimodal composition, and adaptive access. A structured integrative review of 183 works examines when detection evidence supports governance decisions. SCOPE organizes evidence by safety policy, control channel, observation unit, probability regime, and enforcement action. The synthesis identifies channel substitution, repeated-access amplification, and judge dependence as recurring limits on evidence transfer. Worked cases expose unsupported extensions and specify necessary comparisons. Governance claims require evidence matched to deployment conditions and intervention consequences. SCOPE makes these gaps inspectable but requires empirical validation.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Natural language processing**; **Computer vision**; • **Security and privacy** → *Human and societal aspects of security and privacy*; • **Human-centered computing** → *Empirical studies in HCI*.

Additional Key Words and Phrases: toxicity detection for generative AI, content moderation, generative AI safety, guard models, multimodal safety, text-to-image safety, model components, calibration, governance

1 Introduction

Generative AI systems participate in communication by producing responses and images that people may encounter, reuse, and share. When these outputs express harassment, identity attacks, or threats, the system becomes a source of harmful content as well as a means of processing it. Toxicity detection is therefore necessary for identifying harmful exposure, directing moderation, and evaluating whether safeguards work. These decisions also affect legitimate

Authors' Contact Information: Kun Xu, xukun930@nuaa.edu.cn, Nanjing University of Aeronautics and Astronautics, Nanjing, Jiangsu, China; Yushu Zhang, zhangyushu@jxufe.edu.cn, Jiangxi University of Finance and Economics, Nanchang, Jiangxi, China; Xiangli Xiao, xiaoxiangli@nuaa.edu.cn, Jiangxi University of Finance and Economics, Nanchang, Jiangxi, China; Shuren Qi, shurenqi@cityu.edu.hk, City University of Hong Kong, Hong Kong, China; Tao Wang, wangtao21@nuaa.edu.cn, Nanjing University of Aeronautics and Astronautics, Nanjing, Jiangsu, China; Barbara Carminati, barbara.carminati@uninsubria.it, University of Insubria, Varese, Italy; Elena Ferrari, elena.ferrari@uninsubria.it, University of Insubria, Varese, Italy; Yuming Fang, fa0001ng@e.ntu.edu.sg, Jiangxi University of Finance and Economics, Nanchang, Jiangxi, China.

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